

BigQuery Lakehouse & Analytical Storage Architecture

Target Platform: Google Cloud Platform (GCP)

System of Record: ServiceNow ITSM

Version: 2.0 (Production Blueprint)

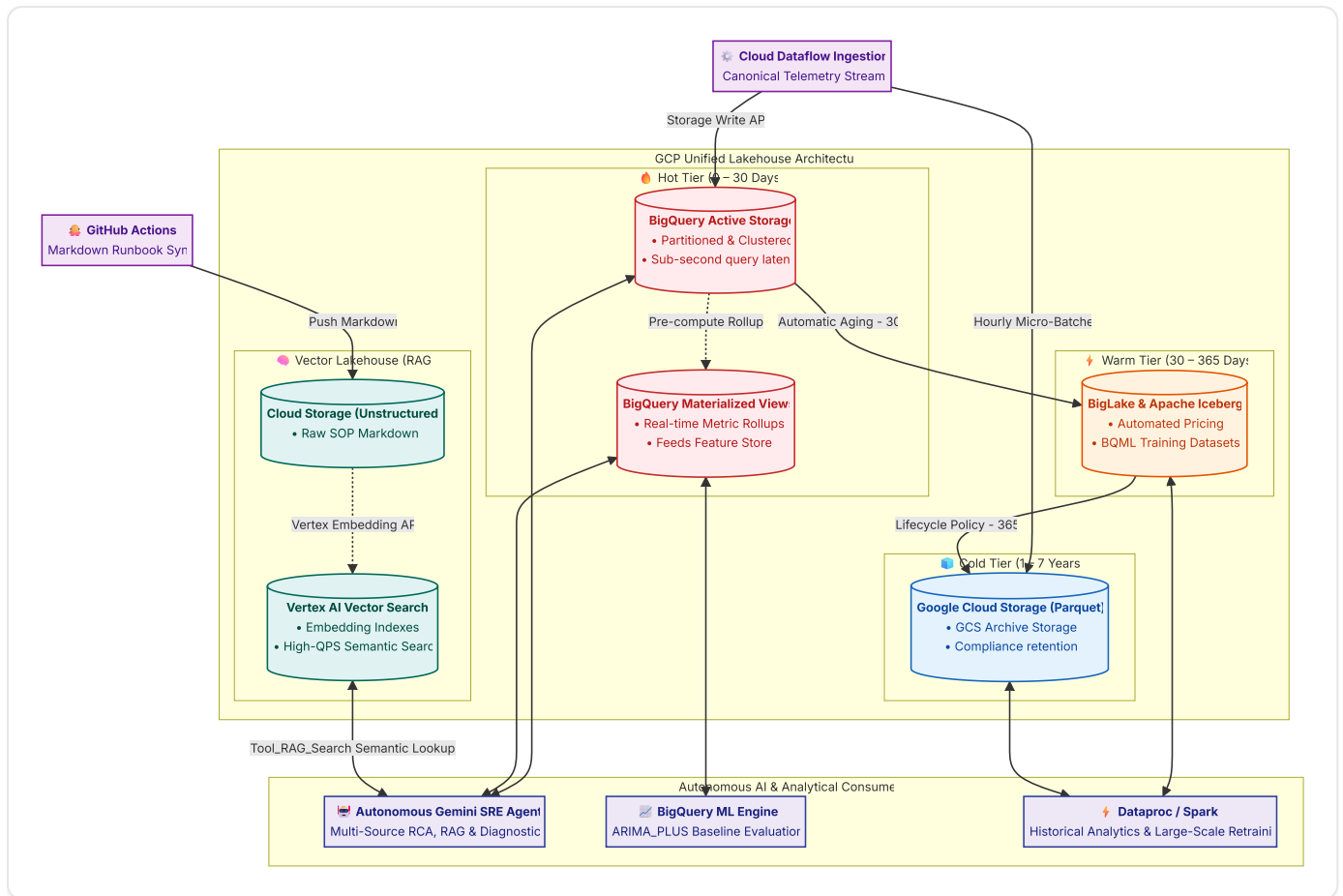
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Unified Lakehouse & Analytical Storage Architecture

1. Overview & Objectives

The **AIOps Data Lakehouse** provides a unified, highly optimized analytical storage layer built natively on **Google Cloud Platform (GCP)**. It serves three distinct workloads essential to the AIOps Intelligence Layer:

- 1. Low-Latency Analytical Querying:** Supporting real-time ML feature extraction, BQML anomaly detection models, and Gemini SRE Agent contextual queries.
- 2. Cost-Effective Long-Term Telemetry Archival:** Storing petabyte-scale historical logs and traces in compressed **Apache Parquet** format for historical trend analysis and compliance auditing.
- 3. Semantic Retrieval-Augmented Generation (RAG):** Storing vectorized representations of Runbooks (SOPs) and historical incident data for real-time retrieval by the Gemini SRE Agent.



2. BigQuery Dataset & Table Design

2.1 Primary Analytical Table: `telemetry_canonical`

- **Table Type:** Partitioned & Clustered Columnar Table.
- **Partitioning Specification:** Partition by day on `DATE(timestamp)`.
 - Partition Expiration: Optional (managed via lifecycle tiering).
- **Clustering Keys:** `source_tool`, `entity.service_name`, `severity`, `domain`.
- **Rationale:** SRE forensic queries and ML anomaly jobs almost always filter by time range, tool, and service name. Clustering reduces bytes scanned by over 85%, ensuring fast query response times and minimal BigQuery query costs.

2.2 Table Schema DDL (BigQuery SQL)

```
CREATE OR REPLACE TABLE `aiops_lakehouse.telemetry_canonical`  
(  
  event_id STRING NOT NULL,  
  timestamp TIMESTAMP NOT NULL,  
  ingestion_timestamp TIMESTAMP NOT NULL,  
  source_tool STRING NOT NULL,  
  domain STRING NOT NULL,  
  severity STRING NOT NULL,  
  entity STRUCT<  
    service_name STRING,  
    environment STRING,  
    host STRING,  
    container_name STRING,  
    cloud_provider STRING,  
    region STRING,  
    cmdb_ci_id STRING  
  >,  
  metrics ARRAY<STRUCT<  
    name STRING,  
    value FLOAT64,  
    unit STRING,  
    dimensions JSON  
  >>,  
  log_payload STRUCT<  
    message STRING,  
    stack_trace STRING,  
    trace_id STRING,  
    span_id STRING,  
    error_code STRING  
  >,  
  business_context STRUCT<  
    orders_per_minute FLOAT64,  
    cart_abandonment_rate FLOAT64,  
    estimated_revenue_loss_usd FLOAT64,  
    funnel_stage STRING,  
    user_cohort STRING  
  >,  
  raw_attributes JSON  
)  
PARTITION BY DATE(timestamp)  
CLUSTER BY source_tool, entity.service_name, severity;
```

2.3 Real-Time AI Feature Store & Materialized Views

To power the `ARIMA_PLUS` models in **BigQuery ML (BQML)** without scanning the massive `telemetry_canonical` table every minute, we utilize **BigQuery Materialized Views**. These act as a continuously updated Feature Store for the ML engine.

For deep-dive specifications on time-series resampling, sliding window alert clustering, trace sessionization, and topology graph construction, see the dedicated [Data Processing & AI Feature Preparation Architecture](#).

By pre-aggregating high-frequency metrics (like Orders Per Minute), the Semantic Router and BQML can evaluate anomalies in sub-seconds with near-zero cost.

```
-- Materialized View for 1-minute OPM (Orders Per Minute) Rollups
CREATE MATERIALIZED VIEW `aiops_lakehouse.telemetry_opm_1min_mv`
PARTITION BY DATE(window_start)
CLUSTER BY service_name
AS
SELECT
  TIMESTAMP_TRUNC(timestamp, MINUTE) AS window_start,
  entity.service_name,
  SUM(business_context.orders_per_minute) AS total_opm,
  AVG(business_context.cart_abandonment_rate) AS avg_abandon_rate
FROM
  `aiops_lakehouse.telemetry_canonical`
WHERE
  source_tool = 'adobe_analytics'
  AND business_context.orders_per_minute IS NOT NULL
GROUP BY
  window_start,
  service_name;
```

3. Storage Tiering & Lifecycle Management

Storage Tier	Data Age	Storage Technology	Storage Cost Profile	Primary Query Engine
Hot Tier	0 – 30 Days	BigQuery Active Storage	Standard BigQuery Active Storage	BigQuery SQL / Vertex AI Agent
Warm Tier	31 – 365 Days	BigLake (Iceberg / BigQuery Long-Term)	50% discount on BigQuery storage pricing	BigQuery SQL / Dataproc Serverless
Cold Tier	1 – 7 Years	Google Cloud Storage (Archive Class)	Ultra-low cost object storage (\$0.0012/GB/mo)	BigQuery BigLake External Queries

4. Vector Lakehouse & RAG Storage

To support the **Context-Aware Incident Response** requirement of the Intelligence Layer, unstructured text must be semantically searchable. This requires a dedicated Vector Lakehouse architecture parallel to the relational BigQuery tables.

4.1 Storage Pipeline

1. **Raw Storage:** Markdown Runbooks (SOPs) are managed in GitHub ("Runbooks as Code"). A GitHub Actions pipeline syncs merged Markdown files directly to a standard **Google Cloud Storage (GCS) Bucket**.
2. **Embedding:** A Cloud Function triggers on new GCS files, chunks the Markdown, and calls the **Vertex AI Text Embeddings API** to generate high-dimensional vectors.
3. **Vector Database:** The resulting vectors, along with metadata (e.g., `service_name: checkout-service`, `author: sre-pod-1`), are indexed in **Vertex AI Vector Search**.

4.2 Retrieval

When the Gemini SRE Agent begins diagnosing a P1 incident, it extracts the `service_name` and `symptoms` from the alert payload. It executes an Approximate Nearest Neighbor (ANN) search against Vertex AI Vector Search to retrieve the exact Markdown SOP required for remediation within milliseconds.

5. Integration with the Autonomous Gemini SRE Agent

The **Autonomous Gemini SRE Agent** relies heavily on the Hot Tier and Materialized Views of this Lakehouse.

When an incident signature is triggered, the Agent uses **Function Calling (Tools)** to query BigQuery securely:

- **Tool_Topology_Graph** : Traverses the `topology_service_graph` table to inspect upstream caller and downstream dependency trees.
- **Tool_Diagnostic_Sandbox** : Executes AST-safe SQL queries against `aiops_lakehouse.telemetry_canonical` to correlate concurrent Splunk error logs and Dynatrace PurePaths matching the incident's `trace_id` (`$\pm 10\text{ minutes}`).
- **Materialized Baseline Queries:** Reads `telemetry_opm_1min_mv` via BigQuery BI Engine in `<500\text{ms}` to verify if technical alerts correspond to active digital revenue drop-offs.

6. High Availability & Disaster Recovery (Resiliency)

To ensure the AIOps platform remains available during regional outages, the storage layer is designed for maximum resilience:

1. **Multi-Region Deployments:** The BigQuery `aiops_lakehouse` dataset and all GCS Cold Tier buckets are configured as **Multi-Region** (e.g., `US` or `EU`). This ensures that if a single GCP region (like `us-central1`) experiences an outage, the SRE team retains access to all historical telemetry and ML models to diagnose the outage.
 2. **Vertex AI High Availability:** The Vector Search index is deployed with multiple replicas across distinct zones within the region to guarantee high QPS availability for the Gemini SRE Agent.
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7. Security, Governance & PII Compliance

Retail telemetry often inadvertently captures sensitive data. The Lakehouse implements strict defense-in-depth:

1. **Cloud DLP Masking:** Before writing to BigQuery, the Dataflow ingestion pipeline utilizes the Cloud Data Loss Prevention (DLP) API to automatically detect and redact PII (e.g., credit card numbers, email addresses) from log payloads.
 2. **Row-Level Security (RLS):** BigQuery Row-Level Security policies are applied to `telemetry_canonical` so that SRE pods can only query telemetry for the specific microservices (e.g., `service_name = 'checkout-service'`) they own.
 3. **Encryption at Rest (CMEK):** All data stored in BigQuery, Cloud Storage, and Vertex AI is encrypted at rest using **Customer-Managed Encryption Keys (CMEK)** via Cloud KMS, fulfilling strict retail compliance mandates.
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8. Performance & Cost Optimization Best Practices

1. **Parquet Columnar Format:** All batched data files written to Cloud Storage utilize **Apache Parquet with Snappy or ZSTD compression**, optimizing storage footprint and scan efficiency.
 2. **Partition Pruning:** All analytical queries generated by the Gemini SRE Agent and Semantic Router must include mandatory time range filters (`WHERE timestamp > TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 1 HOUR)`).
 3. **BigQuery BI Engine:** Enabled with a 50 GB memory reservation to accelerate real-time Materialized View reads and high-frequency anomaly evaluation queries by BQML.
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9. Related Architectural Specifications

- [Data Processing & AI Feature Preparation](#)
 - [AIOps Intelligence Layer Specification](#)
 - [Ingestion Master Blueprint](#)
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